

# Product design Final presentation

## TEAM 6

- Fardin Islam Mahe
- Swann Joly
- Geehun Kong
- Salim Sadman Khan Tasin
- Md Kauser
- Arjun Bhandari



# Design Of Steel Element Packaging Device

The goal of our project is to create a safe and efficient way to package large steel elements used in industrial environments. These steel elements must be protected from rain, moisture, dirt, and damage during transportation and storage.

Currently, packaging takes a lot of time and may cause safety risks. Our task is to design a better solution that is faster, safer, and suitable for production use.

In this presentation, we will explain the problem, our design idea, and how our solution improves the packaging process.



# Current Packaging Challenges & Current Market Solutions

## Challenges

**Safety Risks:** Hazardous "hot-work" required underneath heavy elements using gas flames.

**Efficiency Bottleneck:** Processing time exceeds 1 hour per element, limiting throughput.

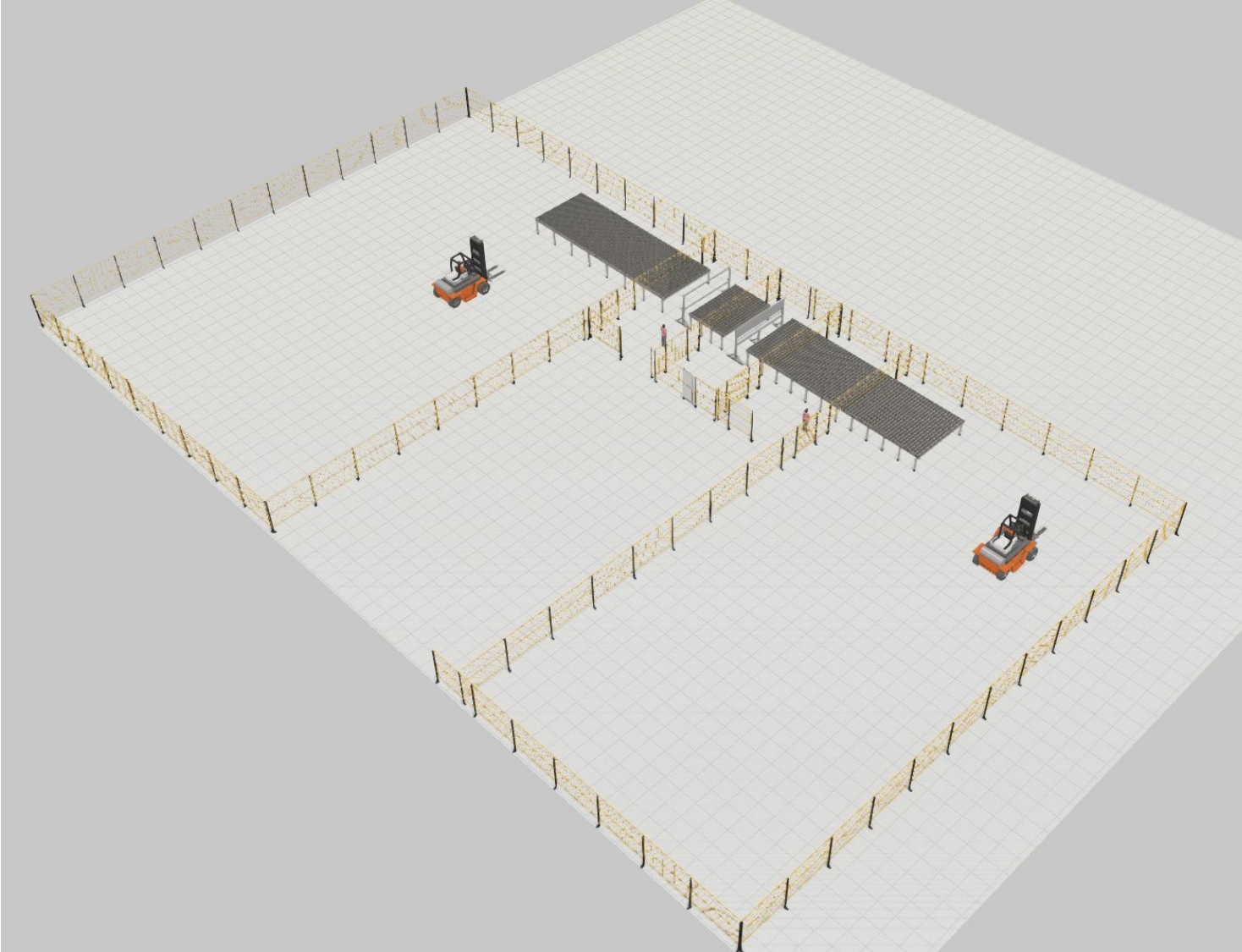
**Size Complexity:** Extreme variability in dimensions (up to 12m) and weights (5,000 kg).



## Existing Solution

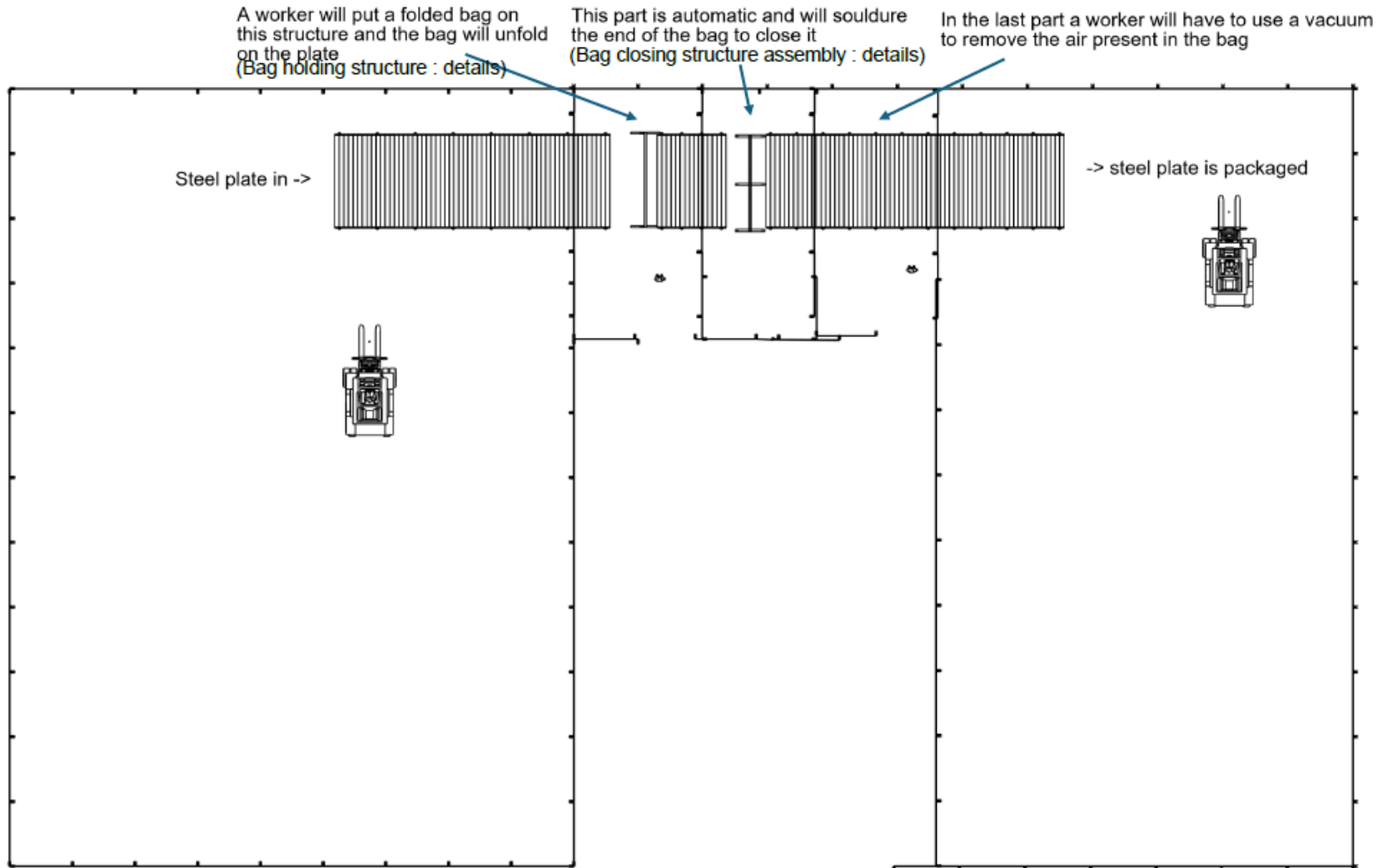
- Manual wrapping with 8m wide shrink-wrap roll material.
- Gas flame heating used to achieve an airtight seal.
- Relatively effective for weather protection and storage.
- **Conclusion:** Incompatible with future serial production goals.

# Our Solution



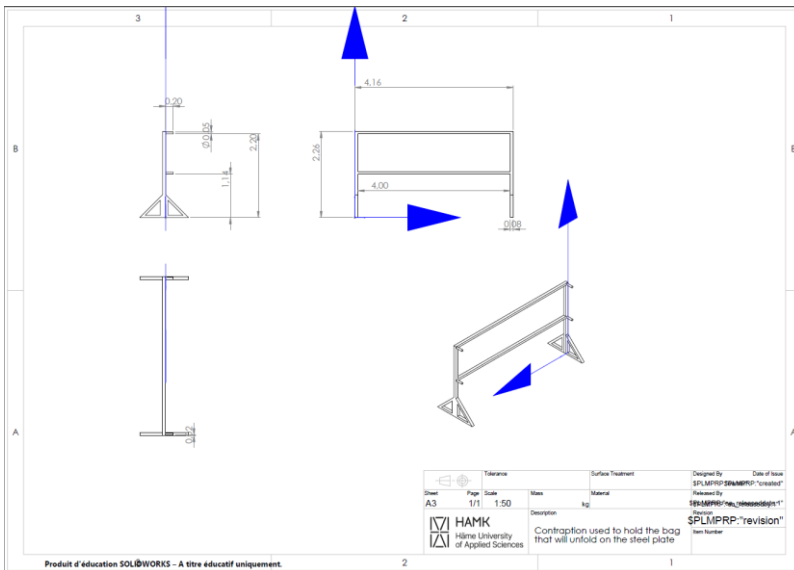
- An automated system designed to vacuum-package large steel plates efficiently and safely.
- Plates are first placed onto a conveyor by a forklift and automatically transported through the system.
- A pre-positioned bag is manually installed at the packaging station. As the plate moves forward, it enters the bag
- The system then seals the bag using a heat-sealing bar and removes the air to create a vacuum.
- Finally, the packaged plate is conveyed to the output area, where it is collected by a forklift for storage or transport.

# Product Overview

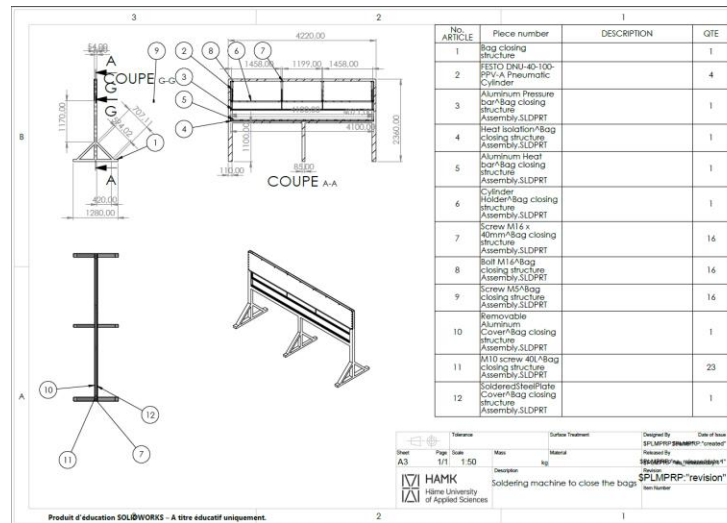


- Our solution will package the steel plates in 18-22 minutes depending on the size of the plates
- The package will be almost airtight to avoid moisture
- The bags used will also protect it against UV using black bags

# Product overview



## Bag holding structure



Automatic soldering station use to close the bag



### Screenshot oh the solidworks assembly

# Main Features

- UV resistant material
- Adjustable frame
- Moisture protection
- Vacuum sealing
- No fire or gas used
- Safe industrial operation



# Risk Assessment

Risk Matrix

| Likelihood \ Severity | Rarely (1) | Unlikely (2) | Possible (3) | Likely (4) | Highly Likely (5) | People                      | Assets          |
|-----------------------|------------|--------------|--------------|------------|-------------------|-----------------------------|-----------------|
| Very High (5)         | 5          | 10           | 15           | 20         | 25                | Death                       | Severe Damage   |
| High (4)              | 4          | 8            | 12           | 16         | 20                | Major Injury                | Major Damage    |
| Medium (3)            | 3          | 6            | 9            | 12         | 15                | Over 3 Day Injury           | Moderate Damage |
| Low (2)               | 2          | 4            | 6            | 8          | 10                | Under 3 Day Injury          | Slight Damage   |
| Very Low (1)          | 1          | 2            | 3            | 4          | 5                 | Minor Injury (No lost Time) | No Damage       |

| Score | Priority | Action                                                                       |
|-------|----------|------------------------------------------------------------------------------|
| 1~4   | Low      | Monitor and maintain existing controls                                       |
| 5~9   | Medium   | Implement additional control measures where practical                        |
| 10~15 | High     | Immediate corrective action required to reduce risk                          |
| 16~25 | Critical | Stop operation immediately until risk is eliminated or significantly reduced |

# Risk Assessment

| # | Hazard Identified               | Risk Description                          | Initial |   | Initial Score (L x S) | Mitigations                                                                                       | Final |   | Final Score (L x S) | Responsible Team       |
|---|---------------------------------|-------------------------------------------|---------|---|-----------------------|---------------------------------------------------------------------------------------------------|-------|---|---------------------|------------------------|
|   |                                 |                                           | L       | S |                       |                                                                                                   | L     | S |                     |                        |
| 1 | Forklift collision or load drop | Severe injury, Crushing, Equipment damage | 2       | 5 | 10                    | Designated forklift operating zones, Proper load securing procedures, Certified operator training | 1     | 5 | 5                   | Logistics Team         |
| 2 | Conveyor pinch/crush points     | Hand entrapment, Crushing injury          | 3       | 4 | 12                    | Physical safety guards, Emergency stop systems, Restricted operator access                        | 1     | 4 | 4                   | Mechanical Design Team |
| 3 | Heat sealing burns              | Thermal burns, Skin injury                | 3       | 3 | 9                     | Thermal insulation barriers, Warning Signage, Operator protective equipment                       | 1     | 3 | 3                   | Process Design Team    |
| 4 | Vacuum sealing failure          | Packaging failure, Transport instability  | 3       | 2 | 6                     | Routine equipment inspections, Seal quality verification, Preventive maintenance schedules        | 1     | 2 | 2                   | Quality Control Team   |
| 5 | Ergonomic strain                | Repetitive strain, Operator fatigue       | 4       | 2 | 8                     | Adjustable workstation design, Task rotation, Ergonomic training                                  | 2     | 2 | 4                   | Operation Team         |

# Cost calculation

## PRODUCT COST & ESTIMATION

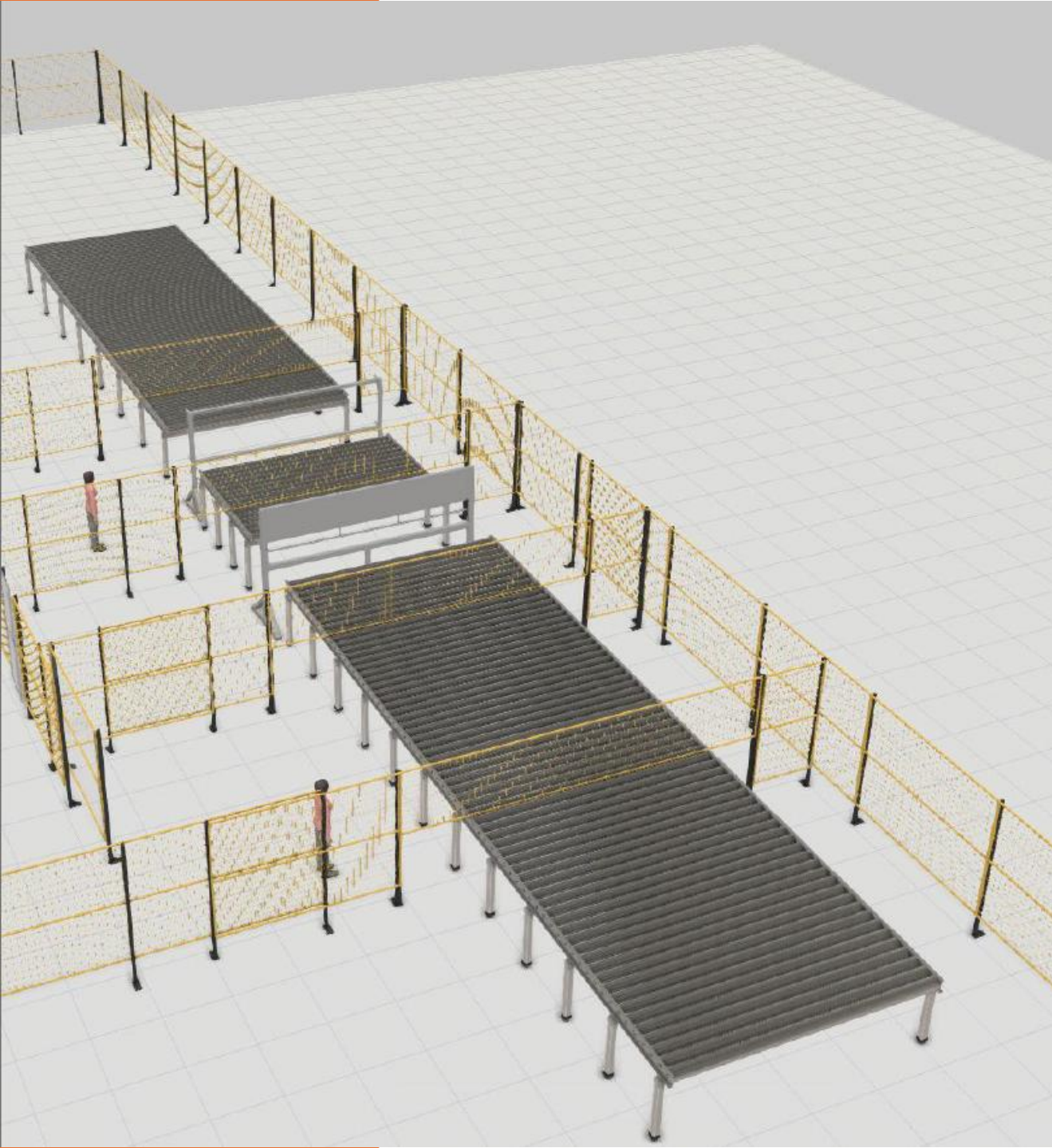
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### TOTAL COST SUMMARY

| Cost Category                         | Amount (€)          |
|---------------------------------------|---------------------|
| A. Structural & Mechanical Components | 3 235,00 €          |
| B. Pneumatic System                   | 640,00 €            |
| C. Fasteners & Assembly Hardware      | 61,35 €             |
| D. Material Handling System           | 215 000,00 €        |
| E. Safety System                      | 22 750,00 €         |
| F. Fabrication Consumables            | 316,00 €            |
| G. Bag Material (Black LDPE – Kivo)   | 1 148,16 €          |
| <b>BASE PROJECT COST</b>              | <b>243 150,51 €</b> |

### FINAL PROJECT ESTIMATE

| Cost Category                        | Amount (€)          |
|--------------------------------------|---------------------|
| <b>Base Material Cost</b>            | <b>243 150,51 €</b> |
| Labor Cost – Low Estimate            | 3 000,00 €          |
| Labor Cost – High Estimate           | 5 000,00 €          |
| Contingency (10% of Base Cost)       | 24 315,05 €         |
| <b>TOTAL ESTIMATED BUDGET – LOW</b>  | <b>270 465,56 €</b> |
| <b>TOTAL ESTIMATED BUDGET – HIGH</b> | <b>272 465,56 €</b> |



## Improvement Over Existing Solution

The proposed solution significantly improves the current manual shrink-wrapping process.

The new system introduces a semi-automated and controlled process.

The use of a vacuum-sealed bag provides better protection against moisture and environmental conditions compared to shrink wrap.

Overall, the new solution is more efficient, safer, and scalable, making it compatible with future serial production requirements.



# Summary

This project focused on the design of an automated system for vacuum packaging large steel plates, ranging from 1×2 m up to 3×12 m and weighing up to 5 tons.

The proposed solution combines forklift handling with an automated conveyor system to transport the plates through different stages: positioning, bag insertion, sealing, and vacuum packaging. A pre-folded bag is manually placed at the packaging station, after which the process is fully automated.

Particular attention was given to the mechanical design, including the structure, the sealing system, and the integration of pneumatic components to ensure reliable and uniform operation over large dimensions.

The system aims to reduce manual work, improve packaging consistency, and protect the plates during storage and transport. Estimated cycle times range from 10 to 20 minutes per plate, depending on size.

Overall, this project demonstrates a realistic and industrially viable solution for handling and packaging large-scale steel components.